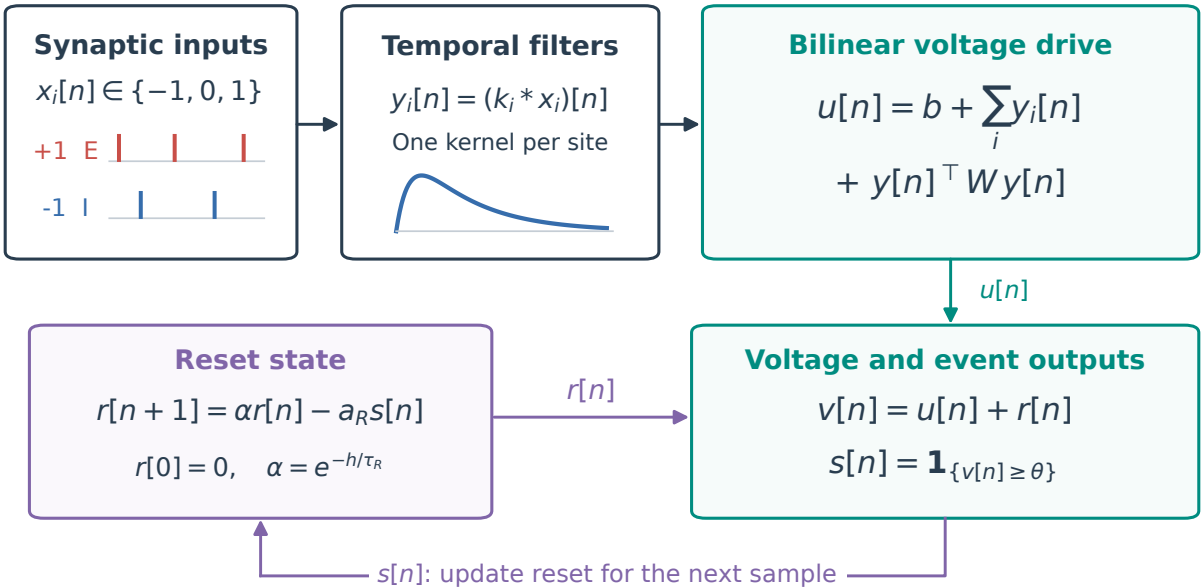
Synaptic inputs
$x_i[n] \in \{-1, 0, 1\}$
+1 E
-1 I
Temporal filters
$y_i[n] = (k_i * x_i)[n]$
One kernel per site
Bilinear voltage drive
$u[n] = b + \sum_i y_i[n]$
$+ y[n]^\top W y[n]$
$u[n]$
Reset state
$r[n+1] = \alpha r[n] - a_R s[n]$
$r[0] = 0, \quad \alpha = e^{-h/\tau_R}$
$r[n]$
Voltage and event outputs
$v[n] = u[n] + r[n]$
$s[n] = \mathbf{1}_{\{v[n] \geq \theta\}}$
$s[n]$: update reset for the next sample